



Solutions

Category A

Problem 1.

Aman and Berdi, two biologists, they invented a new type of bacteria such that they can control the division of this bacteria into several parts. They are also participants of *OMOUS-2025*, with the aim to train for the first problem of *OMOUS-2025*. They play the following game:

Initially, they take 1 bacterium and take some natural number n . At each turn, a player selects any number k from 1 to n . Then the player divides each bacterium into k part. Once selected k cannot be selected twice. If the number of the population of bacterium's is divisible by n after the turn of any player, that player loses.

Determine who has a winning strategy depending on the given number if Aman starts first n .

Solution:

Let's analyze the following cases.

If $n = 4k$ then Aman can choose the number $2k$. Then the one who chooses the even number first loses. There are only $2k$ odd numbers left. So, after k moves, the odd numbers will be exhausted, and Berdi will have to choose an even number first, and therefore he will lose.

If $n = 4k + 2$ then Aman can write the number $2k + 1$. Then, again, the one who first chooses an even number loses. There are only $2k$ odd numbers left, so Berdi also loses.

If n is odd and divisible by p^2 , where p is a prime number, then Aman can write the number $\frac{n}{p}$. The loser is the one who first chooses a number divisible by p . There are only $\frac{n}{p} - 1$ such numbers: even number, since one such number $\frac{n}{p}$ has already been chosen. Then there are an even number of valid numbers left. Therefore, Berdi also loses.

The last case is if n is the product of several (possibly one) pairwise distinct odd primes: say, $n = p_1 p_2 \dots p_k$. Then Berdi follows the following strategy. Before his move, if Aman has not yet lost after his last move, then the product of the numbers chosen at the moment is not divisible by some p_i . Thus, none of the currently chosen numbers are multiples of p_i , and their total number $\frac{n}{p_i}$, odd. Therefore, there is only an even number of numbers not divisible by p_i : $n - \frac{n}{p_i}$. Only such numbers were chosen for dividing and Aman was the first to choose them, so there will always be an unchosen number of Berdi, because the number of numbers already chosen before his move is always odd. So, Berdi can always make a move that does not lead to defeat, and therefore he will win, because the game must obviously end in defeat for one of the players.

Marking scheme problem 1 (Cat A)

2 points	Solving for $n=4k$
2 points	Solving for $n=4k+2$
3 points	Solving for n that have prime factors with degrees greater than 1
3 points	Solving for n that have prime factors with degrees equal to 1



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Problem 2

Compute

$$\prod_{n=1}^{\infty} \frac{(2n)^4 - 1}{(2n+1)^4 - 1} \frac{n^2}{(n+1)^2}.$$

Solution:

Notice that

$$\begin{aligned} P &:= \prod_{n=1}^{\infty} \frac{(2n)^4 - 1}{(2n+1)^4 - 1} \frac{n^2}{(n+1)^2} = \prod_{n=1}^{\infty} \frac{(2n)^2 - 1}{(2n+1)^2 - 1} \frac{n^2}{(n+1)^2} \cdot \prod_{n=1}^{\infty} \frac{(2n)^2 + 1}{(2n+1)^2 + 1} \\ &= \prod_{n=1}^{\infty} \frac{(2n-1)(2n+1)}{(2n+2)2n} \frac{n^2}{(n+1)^2} \frac{(2n)^2}{(2n+1)^2} \cdot \prod_{n=1}^{\infty} \frac{1 + \frac{1}{(2n)^2}}{1 + \frac{1}{(2n+1)^2}} = P_1 P_2. \end{aligned}$$

Clearly

$$P_1 = \prod_{n=1}^{\infty} \frac{(2n-1)(2n+1)}{(2n+2)2n} \frac{n^2}{(n+1)^2} \frac{(2n)^2}{(2n+1)^2} = \lim_{N \rightarrow \infty} \frac{2}{(2N+2)(2N+1)} \frac{1}{(N+1)^2} = 0$$

Now let's show that P_2 is convergent. It is easy to notice that

$$P_2 = \prod_{n=1}^{\infty} \frac{1 + \frac{1}{(2n)^2}}{1 + \frac{1}{(2n+1)^2}} = \frac{\prod_{n=1}^{\infty} \left(1 + \frac{1}{(2n)^2}\right)^2}{\prod_{n=1}^{\infty} \left(1 + \frac{1}{(n)^2}\right)}.$$

Using the well-known formula

$$\frac{\sin x}{x} = \prod_{n=1}^{\infty} \left(1 - \frac{x^2}{\pi^2 n^2}\right),$$

we get

$$\prod_{n=1}^{\infty} \left(1 + \frac{1}{n^2}\right) = \frac{\sin(\pi i)}{\pi i} = \frac{\sinh(\pi)}{\pi}, \quad \prod_{n=1}^{\infty} \left(1 + \frac{1}{(2n)^2}\right) = \frac{\sin\left(\frac{\pi}{2}i\right)}{\frac{\pi}{2}i} = \frac{2\sinh\left(\frac{\pi}{2}\right)}{\pi}.$$

Therefore $P_2 = \frac{2}{\pi} \operatorname{th}\left(\frac{\pi}{2}\right)$ and $P = 0$.

Marking Scheme For Problem 2(Cat A)

2 points	For separating and $P_1 = 0$
1 points	For note that $\frac{\sin x}{x} = \prod_{n=1}^{\infty} \left(1 - \frac{x^2}{\pi^2 n^2}\right)$,
7 points	For calculating $P_2 = \frac{2}{\pi} \operatorname{th}\left(\frac{\pi}{2}\right)$



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Marking Scheme for Problem 3 (Cat A)

2P	Using or mentioning of possibility of usage of cosine theorem
1P	Several using of cosine theorem
1P	Some transformations of formulas (3) and (4)
1P	Derivation of the formula $8S_{123}S_{146}\cos\varphi = 4S_{123}^2 + 4S_{146}^2 + 2X$
1P	Some transformations of formulas (1) and (2)
1P	Application of the Heron formula
3P	Completion of the solution



Problem 3

Let an arbitrary tetrahedron be given with pairwise intersecting edges of length l_1 and l_5 , l_2 and l_6 , l_3 and l_4 , respectively. Show that the cosine of the dihedral angle φ at edge l_1 can be expressed through the areas of adjacent faces S_{123} and S_{146} , as well as the squares of the edge lengths using the formula

$$\cos \varphi = \frac{4S_{123}^2 + 4S_{146}^2 - l_1^2 l_5^2 + \frac{1}{4} ((l_3^2 + l_4^2) - (l_2^2 + l_6^2))^2}{8S_{123} \cdot S_{146}}$$

Solution:

Запишем теорему косинусов для каждой грани тетраэдра:

$$S_{123}^2 = S_{146}^2 + S_{245}^2 + S_{356}^2 - 2S_{146}S_{245} \cos l_4 - 2S_{146}S_{356} \cos l_6 - 2S_{245}S_{356} \cos l_5, \quad (1)$$

$$S_{146}^2 = S_{123}^2 + S_{245}^2 + S_{356}^2 - 2S_{123}S_{245} \cos l_2 - 2S_{123}S_{356} \cos l_3 - 2S_{245}S_{356} \cos l_5, \quad (2)$$

$$S_{245}^2 = S_{123}^2 + S_{146}^2 + S_{356}^2 - 2S_{123}S_{146} \cos \varphi - 2S_{123}S_{356} \cos l_3 - 2S_{146}S_{356} \cos l_6, \quad (3)$$

$$S_{356}^2 = S_{123}^2 + S_{146}^2 + S_{245}^2 - 2S_{123}S_{146} \cos \varphi - 2S_{123}S_{245} \cos l_2 - 2S_{146}S_{245} \cos l_4, \quad (4)$$

где S_{ijk} — площадь грани, ограниченной ребрами l_i, l_j, l_k , а $\cos l_i$ — косинус двугранного угла при ребре l_i .

Складывая (3) и (4), получаем:

$$8S_{123}S_{146} \cos \varphi = 4S_{123}^2 + 4S_{146}^2 + 2X,$$

где

$$X = -2S_{123}S_{356} \cos l_3 - 2S_{146}S_{356} \cos l_6 - 2S_{123}S_{245} \cos l_2 - 2S_{146}S_{245} \cos l_4.$$

В свою очередь, складывая формулы (1) и (2), находим

$$2X = 8S_{245}S_{356} \cos l_5 - 4S_{245}^2 - 4S_{356}^2.$$

Выражая S_{245}^2 и S_{356}^2 по формуле Герона (через квадраты длин ребер), а $8S_{245}S_{356} \cos l_5$ через скалярное произведение векторных произведений (и применяя соответствующую формулу, а затем теорему косинусов)

$$\begin{aligned} 8S_{245}S_{356} \cos l_5 &= -2[\vec{l}_5 \times \vec{l}_3][\vec{l}_2 \times \vec{l}_5] = \\ &= \frac{1}{2} \begin{vmatrix} 2\vec{l}_5^2 & 2\vec{l}_5 \vec{l}_2 \\ 2\vec{l}_3 \vec{l}_5 & 2\vec{l}_3 \vec{l}_2 \end{vmatrix} = \frac{1}{2} \begin{vmatrix} 2l_5^2 & l_2^2 + l_5^2 - l_4^2 \\ l_3^2 + l_5^2 - l_6^2 & l_2^2 + l_3^2 - l_1^2 \end{vmatrix}, \end{aligned}$$

мы получаем явное выражение $2X$ через квадраты длин ребер. Наконец, подставляя найденное выражение для $2X$ в формулу (5), после приведений подобных членов приходим к требуемой формуле.



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Problem 4

Prove that

$$\text{rank} M + \text{rank}(M^{2023} - M^{2025}) = \text{rank}(M - M^2) + \text{rank}(M^{2023} + M^{2024})$$

for all $M \in M_n(C)$.

Solution:

It is known that for any $A, B \in M_n(C)$:

$$\text{Ker} A \subseteq \text{Im} B \Leftrightarrow \text{rank} A + \text{rank} B = n + \text{rank}(AB) \quad (1)$$

(this is the equality case in Sylvester's inequality for ranks).

First, we claim that $\text{Ker}(I_n - M) \subseteq \text{Im}(M^{2023} + M^{2024})$. Indeed,

$$X \in \text{Ker}(I_n - M) \Leftrightarrow MX = X \Rightarrow M^{2023}X = X \Rightarrow M^{2024}X = X \Rightarrow (M^{2023} + M^{2024})\left(\frac{1}{2}X\right) = X \Rightarrow X \in \text{Im}(M^{2023} + M^{2024}).$$

Then, according to (1),

$$\text{rank}(I_n - M) + \text{rank}(M^{2023} + M^{2024}) = n + \text{rank}(I_n - M)(M^{2023} + M^{2024}),$$

hence

$$n + \text{rank}(M^{2023} - M^{2025}) = \text{rank}(I_n - M) + \text{rank}(M^{2023} + M^{2025}). \quad (2)$$

Also, $\text{Ker} M \subseteq \text{Im}(I_n - M)$, since

$$X \in \text{Ker} M \Leftrightarrow MX = 0_n \Rightarrow X - MX = X \Rightarrow (I_n - M)X = X \Rightarrow X \in \text{Im}(I_n - M),$$

then, by (1),

$$\text{rank} M + \text{rank}(I_n - M) = n + \text{rank}(M - M^2). \quad (3)$$

By adding (2) and (3), we arrive to the conclusion.

Marking Scheme for Problem 4(Cat A)

1 point	For equality case of Sylvester's inequality
2 point	Justification of $X \in \text{Im}(M^{2023} + M^{2024})$.
2 point	Using (1) and getting (2)
2 point	Justification $\text{Ker} M \subseteq \text{Im}(I_n - M)$.
2 point	Using (1) and getting (3)
1 point	Getting given equality using (2) and (3)



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Problem 5

Let n and T be positive integers. Bob has $8n$ stones with weights $1, 2, \dots, 8n$. The stones are placed on the pans of a balance scale, each stone is either on the left or the right pan, so that the total weight on both sides is equal. Bob is allowed to move one stone at a time from one side of the scale to the other side. However, if at any moment the absolute difference in total weights of the two pans exceeds T , the scale breaks. Determine all positive integers T such that it is always possible to eventually move every stone to the opposite pan (i.e., every stone ends up on the other side from where it started), without ever breaking the scale.

Solution:

Answer: $T \geq 8n$

We must have $T \geq 8n$, as otherwise we can never move the stone of weight $8n$. We will show that $T = 8n$ by showing that, for any initial configuration, there is a sequence of moves, never increasing the absolute value of the difference above $8n$, that results in every stone ending up on the opposite side of the scale. Because moves are reversible, it suffices to do the following: exhibit at least one configuration C for which this can be achieved, and show that any initial configuration can reach such a configuration C by some sequence of moves.

Consider partitioning the weights into pairs $(t, 8n+1-t)$. Suppose that each side of the balance contains n of those pairs. If one side of the balance contains the pair $(t, 8n+1-t)$ for $1 \leq t \leq 4n$ and the other side contains $(4n, 4n+1)$, then the following sequence of moves swaps those pairs between the sides without ever increasing the absolute value of the difference above $4n$.

$$t, 8n+1-t \quad | \quad 4n, 4n+1 \quad (1)$$

$$t, 4n, 8n+1-t \quad | \quad 2n+1 \quad (2)$$

$$t, 4n \quad | \quad 4n+1, 8n+1-t \quad (3)$$

$$t, 4n, 4n+1 \quad | \quad 8n+1-t \quad (4)$$

$$4n, 4n+1 \quad | \quad t, 8n+1-t \quad (5)$$

Applying this sequence twice swaps any two pairs $(t, 8n+1-t)$ and $(t', 8n+1-t')$ between the sides. So we can achieve an arbitrary exchange of pairs between the sides, and C can be any configuration where each side of the balance contains n of those pairs.

We now show that any initial configuration can reach one where each side has n of those pairs. Consider a configuration where one side has total weight $A-s$ and the other has total weight $A+s$, for some $0 \leq s \leq 4n$, and where some pair is split between the two sides. (If no pair is split between the two sides, they must have equal weights and we are done.) Valid moves include moving any weight w with $1 \leq w \leq 4n+s$ from the $A+s$ side to the $A-s$ side, and moving any weight w with $1 \leq w \leq 4n-s$ from the $A-s$ side to the $A+s$ side. Suppose the pair $(t, 8n+1-t)$, with $t \leq 4n$, is split between the sides. If t is on the $A+s$ side, or on the $A-s$ side and $t \leq 4n-s$, it can be moved to the other side. Otherwise, t is on the $A-s$ side and $t \geq 4n-s+1$, so $8n+1-t \leq 4n+s$ is on the $A+s$ side and can be moved to the other side. So we can unite the two weights from that pair without splitting any other pair, and repeating this we reach a configuration where no pair is split between the sides.

Marking Scheme for Problem 5(Cat A)

1 point	Claiming answer: $T \geq 8n$
3 points	For partitionings (1), (2), (3), (4), (5)
6 points	Other remained part



Problem 6.

Let $A = (a_{ij})_{i,j=1}^n \in M_n(R)$ be a positive semi-definite matrix. Prove that the matrix $B = (b_{ij})_{i,j=1}^n$, where $b_{ij} = \arcsin(x^{i+j}) \cdot a_{ij}$, is also positive semi-definite for all $x \in (0, 1)$.

Solution.

Let's take $f(x) = \arcsin(x)$ (1). We begin with the following two lemmas.

Lemma 1: Let us consider the function (1), we have

$$f^{(n)}(x) = \frac{Q_{n-1}(x)}{(1-x^2)^{2n-1/2}} \quad (n = 1, 2, 3, \dots)$$

$Q_{n-1}(x)$ is a polynomial of integer coefficients. These polynomials can be obtained using the recursive formulae

$$Q_0(x) = 1, Q_n(x) = Q'_{n-1}(x)(1-x^2) + (2n-1)xQ_{n-1}(x) \quad (2) \quad (n = 1, 2, 3, \dots)$$

For these polynomials holds inequality $0 \leq Q_n(x) \leq (n+1) \cdot 2^n$ for $x \in [0, 1]$

Proof of Lemma 1: Above properties are obvious from the induction.

Lemma 2: Let $p_n(x)$ be the n -th Taylor polynomial of $f(x)$ about 0. Then, p_n has non-negative coefficients and $p_n(x) \rightarrow f(x)$, uniformly on any interval $[0, y] \subseteq (0, 1)$.

Proof of Lemma 2: We know that $p_n(x) = \sum_{k=0}^n \frac{f^{(k)}(0)}{k!} \cdot x^k$. Because $f^{(k)}(0) > 0$, the fact that p_n has nonnegative coefficients is obvious. Also, using the integral form of the Taylor remainder, one can deduce that for all $x \in [0, y]$, $|R_{n,f,0}(x)| = \left| \frac{1}{n!} \int_0^x f^{(n+1)}(t)(x-t)^n dt \right| \leq \frac{1}{n!} \int_0^x \frac{Q_n(t)}{(1-t^2)^{2n+1/2}} (x-t)^n dt \leq$

$$\frac{(n+1)2^n}{n!(1-x^2)^{2n+1/2}} \leq \frac{(n+1)2^n}{n!(1-(\frac{3}{2})x^2)^{2n+1/2}} \leq \frac{n+1}{n} \cdot \frac{\frac{2^n}{n!}}{\frac{1}{(n-1)!} \left(2 - \frac{3}{2n}x^2\right)} \xrightarrow{n \rightarrow \infty} 0.$$

Lemma 3: For all $m \in N$, the matrix $C_m = (x^{m(i+j)} a_{ij})_{i,j=1}^n$ is positive semi-definite.

Proof of Lemma 3: For all $y = (y_1 y_2 \dots y_n)^T \in R^n$, we have

$$\begin{aligned} \langle C_m y, y \rangle &= \sum_{i=1}^n a_{ii} x^{2mi} y_i^2 + \sum_{1 \leq i < j \leq n} (a_{ij} + a_{ji}) x^{m(i+j)} y_i y_j \\ &= \sum_{i=1}^n a_{ii} (x^{mi} y_i)^2 + \sum_{1 \leq i < j \leq n} (a_{ij} + a_{ji}) (x^{mi} y_i) \cdot (x^{mj} y_j) \geq 0, \end{aligned}$$

because A is positive semi-definite.

Now, let $\varepsilon > 0$ and $N \in N$ such that $|f(y) - p_N(y)| < \varepsilon$, for all $y \in [0, x]$ (note that $x^{i+j} \in [0, x]$, for all i, j). Also, set $M = \max_{i,j} |a_{ij}|$. Observe that the matrix $B_N = \sum_{k=0}^N \frac{f^{(k)}(0)}{k!} C_k$ is positive semi-definite, as a positive linear combination of positive semi-definite matrices, and that the matrix $B - B_N = (s_{ij})_{i,j=1}^n$, has entries with $|s_{ij}| < M\varepsilon$. Now, fix $y = (y_1 y_2 \dots y_n)^T \in R^n$. Then

$$|(B - B_N)y, y| = \left| \sum_{i=1}^n s_{ii} y_i^2 + \sum_{1 \leq i < j \leq n} (s_{ij} + s_{ji}) y_i y_j \right| \leq M\varepsilon \sum_{i=1}^n |y_i|^2 + 2M\varepsilon \sum_{1 \leq i < j \leq n} |y_i| |y_j| > M\varepsilon \left(\sum_{i=1}^n |y_i| \right)^2,$$

which implies

$$\langle By, y \rangle \geq \langle B_N y, y \rangle - M\varepsilon \left(\sum_{i=1}^n |y_i| \right)^2 \geq -M\varepsilon \left(\sum_{i=1}^n |y_i| \right)^2.$$

Since ε can be arbitrary, we conclude that $\langle By, y \rangle \geq 0$, which proves the desired result.

Marking Scheme for Problem 6(Cat A)

3 points	For Lemma 1 fully completed
3 points	For Lemma 2 completed
4 points	For Lemma 3 completed